

Endocrine system

- secretory cells
- Endocrine glands located in various organs
- Also adipose tissue, connective tissue, nervous tissue and muscle tissue produces hormones
- **DNES** : diffuse neuroendocrine system - unicellular glands in epithelium (GIT, respiratory, kidney ..)
- Products excreted **directly into bloodstream or tissue fluid** not through ducts
- Produce **hormones**
- Intensely vascularised and innervated
- Type of glands
 - **Trabecular**
 - Secretory cells arranged in irregular **chords - forming nests**
 - Hypophysis
 - Adrenal glands
 - **Follicular**
 - Secretory cells arranged in circular structures - **follicles**
 - Thyroid

Endocrine Glands

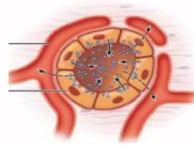
Cord & Clump Type



Secretions stored inside the cells
"Intra-cellular method"

E.g. Most endocrine glands
Pituitary, Adrenal, Parathyroid, Pineal

Follicular Type



Secretions stored outside the cells inside follicles
"Extra-cellular method"

E.g.
Thyroid Gland

HYPOPHYSIS (PITUITARY GLAND)

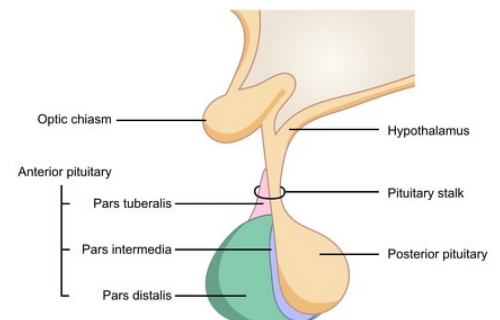
- in sella turcica of sphenoid bone
- Connected with hypothalamus via infundibulum
- 2 parts
 - Adenohypophysis
 - Ectodermal diverticulum (pouch)= Rathkes pouch of stomodeal roof - primitive oral cavity
 - Neurohypophysis
 - Neuroectoderm diverticulum of diencephalic floor (neural connection with brain)

Adenohypophysis

- **Trabecular type of gland** (even though pars intermedia contains follicles)

macroscopic structure

- **Pars tuberalis** - funnel shaped region surrounding infundibulum
- **Pars intermedia** - on interface with neurohypophysis (remnant of Rathkes pouch)
- **Pars distalis** - main part in ventral position (75%)



- Cells

chromophobes :

- do not stain intensely
- Smaller
- Few - no secretory granules
- Less cytoplasm than chromophils
- Undifferentiated cells / already used cells - de granulated

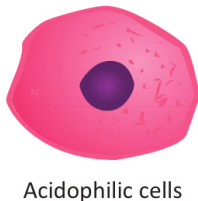


chromophils

- Half of cells of Adenohypophysis
- Divided based on cytoplasmic staining due to secretory granules
 - **Basophilic cells :**
 - **acidophilic cells**
- **Basophilic cells :** acting on glands
 - **Gonadotropic cells :**
 - FSH - follicular stimulating hormone
 - LH - lutenising hormone
 - **thyrotrophic cells :**
 - TSH - Thyrotropin
 - **corticotrophic cells :**
 - ACTH- Adrenocorticotrophic hormone



- **Acidophilic cells :** acting directly on body
 - **Somatotropic cells**
 - STH - somatotropin (growth hormone)
 - **mammotrophic (lactotrophic) cells :**
 - LTH - leutotropic hormone (prolactin)



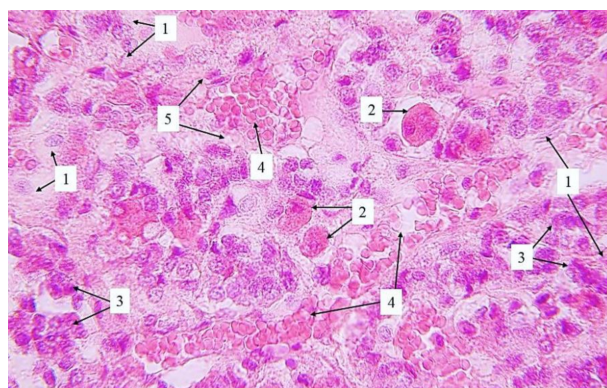
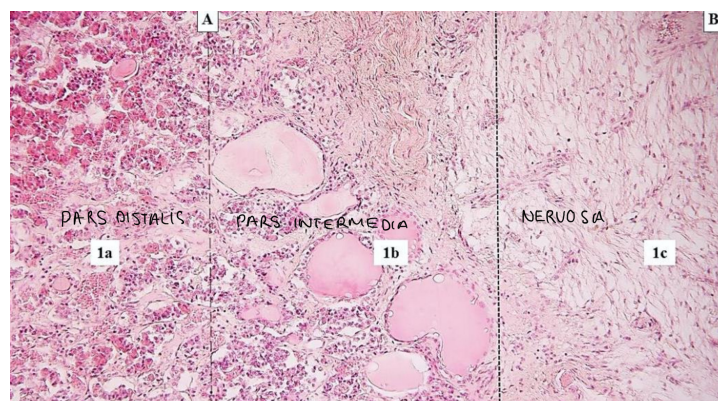
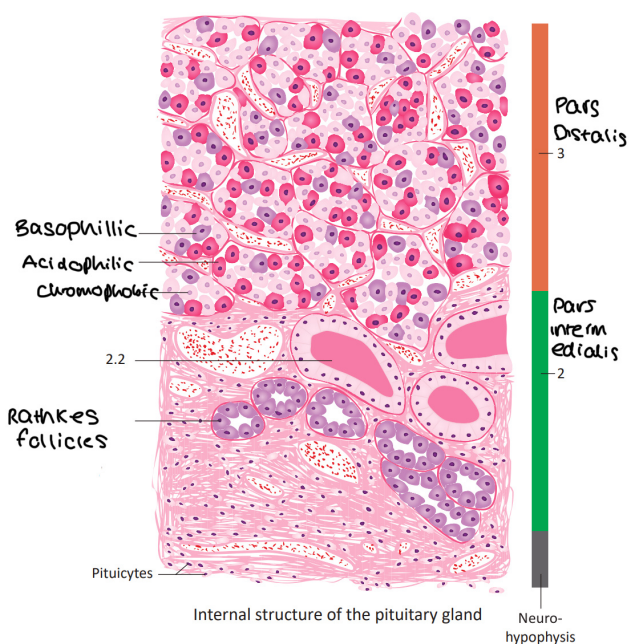
Folliculo-stellate cells

- not hormone producing cells
- Multiple cytoplasmic protrusions
 - Forming network between which endocrine cells are found
- Supporting cells

• microscopic structure

Note : blood supply is via portal circulation

- **Capsule** : over whole hypophysis (anterior and posterior)
 - Connective tissue
 - Trabeculae (vessels and nerves)
- **Pars tuberalis**
 - **Cords / clusters of** glandular cells in close association to vessels
 - Highly vascularised
 - Veins of hypothalamic-hypophyseal system (primary capillary network)
- **Pars intermedia**
 - **Follicles** (of Rathke) lined by cuboidal epithelium filled with colloid
 - Cuboidal epithelium : mainly basophilic (+ chromophobic cells)
- **Pars distalis**
 - **Cords / clusters** of glandular epithelium in close relation to
 - Sinusoid / fenestrated capillaries (secondary capillary network)
 - Fibro-Stellate cells also present
 - **Somatotropic cells** = 50% (acidophilic) - growth hormones
 - Mammatropic cells = ~20% (acidophilic) - prolactin
 - Corticotropic cells = ~15% (basophilic) - adrenocorticotrophic hormone
 - Gonadotropic cells = ~10% (basophils) - FSH / LH
 - Thyrotropic cells = ~5% (basophils) - TSH



Pars distalis – endocrine cells

- 1- chromophobic cells,
- 2- acidophilic cells,
- 3- basophilic cells,
- 4- blood capillaries
- 5- thin connective tissue trabeculae

(staining method: H-E)

Neurohypophysis

- macroscopic structure

- **Pars nervosa** - unmyelinated axons
- **Infundibulum** - neural stalk

- Microscopic structure

- **No secretory cells**
- No hormones **produced** but hormones **stored**

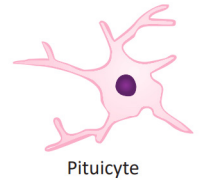
- pars nervosa

- Unmyelinated axons (of neurosecretory neurons)

- Cell bodies of these axons are in the nuclei of the anterior hypothalamus
- **Herrings bodies**
 - Enlarged parts of the unmyelinated axons
 - Accumulation of hormones
 - Hormones released from these vesicles into neurohypophysis/ capillaries
 - **Oxytocin**
 - **Vasopressin (antidiuretic hormone - ADH)**

- Pituicyte

- Specialised glial cell similar to astrocyte
- Irregular with multiple protrusions
- Form 3D network around axonal endings
- Contain lipofucin

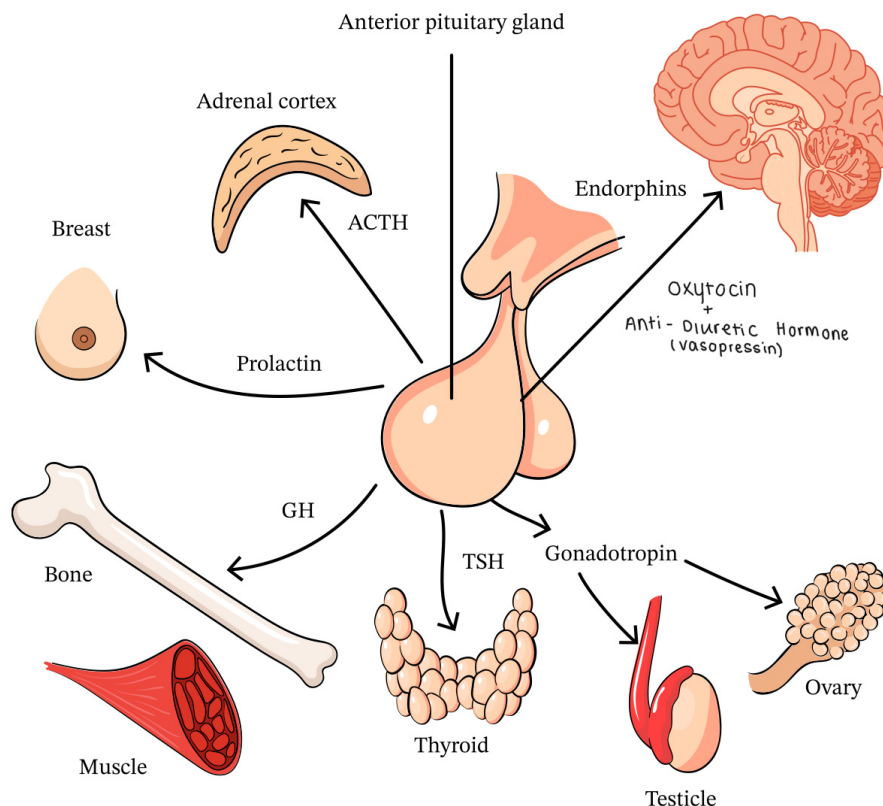


- blood capillaries

- Fenestrated capillaries

- infundibulum

- Bundles of unmyelinated axons



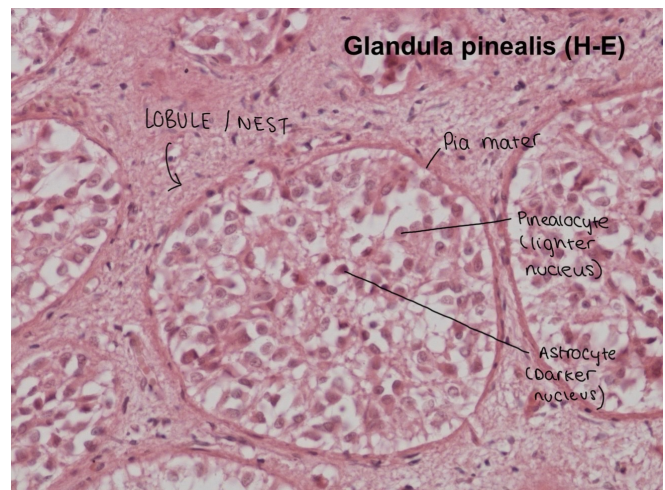
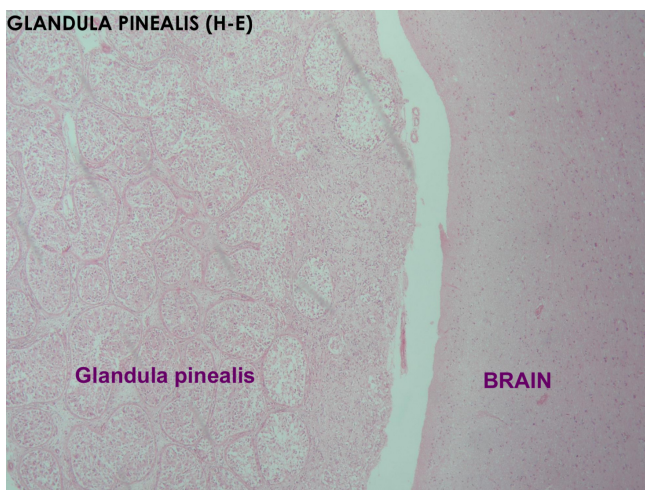
PINEAL GLAND

- unpaired, small pine cone shaped organ
- Remains attached to the brain by a short stalk
- Develops from neuroectoderm in the posterior part of the roof of the diencephalon
- Located in superior posterior wall of the 3rd ventricle
- Cord like arrangement

microscopic structure

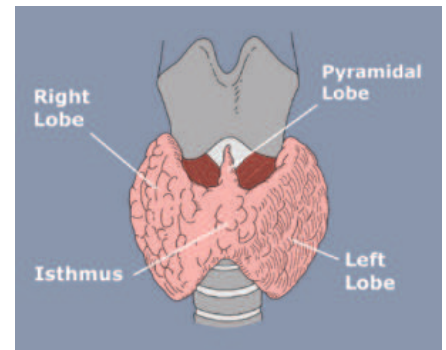
- **pia mater** :
 - Fibrous capsule producing **septa** within the parenchyma dividing it into lobules
 - Septa contain blood vessels
- **parenchyma** :
 - 2 cell types
 - **Pinealocyte**
 - Main neurosecretory cell
 - **Produce melatonin : regulates circadian rhythm**
 - Form clusters / fascicles / nets in the lobules separated by fibrous septa of pia mater
 - Large protrusions which are in contact with blood capillaries
 - Cytoplasm contains lipid droplets
 - **interstitial glial cells - astrocytes**
 - Form 3D network where pinealocytes are located
- **stroma**
 - Reticular connective tissue between the cells containing brain sand
 - Unmyelinated axons and capillaries also surround the parenchyma
- **brain sand (averculus)**
 - Extracellular calcified bodies

Note : pineal gland is sensitive to light, meaning light will suppress secretion of melatonin and darkness will allow secretion of melatonin thus regulating the circadian rhythm



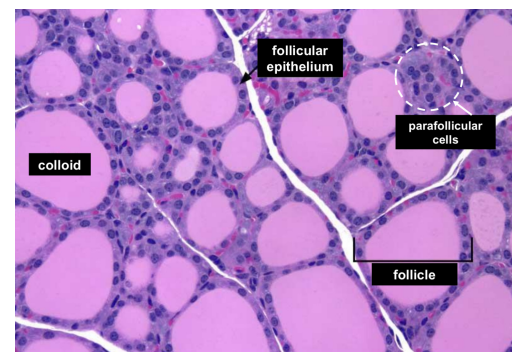
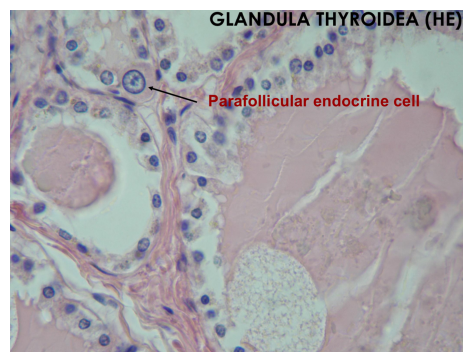
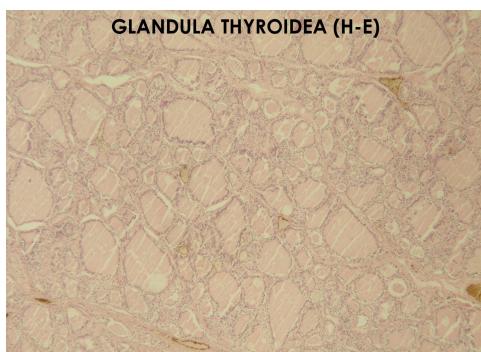
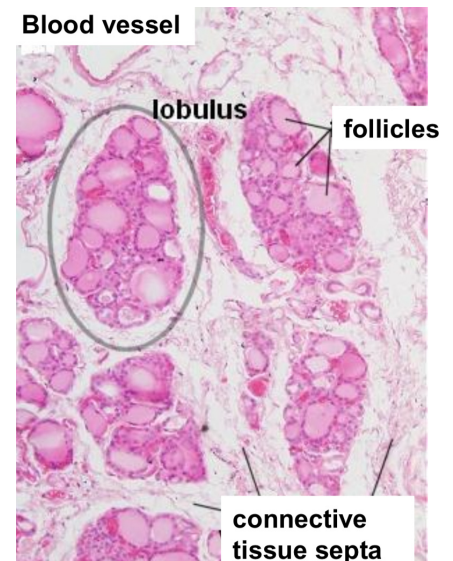
THYROID

- located in anterior neck : adjacent to larynx and trachea
- Consists of
 - **2 lobes** connected together by ..
 - **Isthmus**
 - **Pyramidal lobe** extending up from isthmus
- **follicular type of gland**



Microscopic structure

- **fibrous capsule**
 - Dense CT Containing **septa** dividing the parenchyma into lobules
- **parenchyma**
 - **Follicles** : basic structural and functional unit of the gland
 - Epithelium : **follicular cells**
 - Filled with **colloid**
 - Gel like substance - inactive storage form of the thyroid hormones
 - **Cells :**
 - **Follicular cells**
 - Produce
 - **T3 : triiodothyronine**
 - **T4 : thyroxine**
 - Control basal metabolism
 - Controlled by TSH from Adenohypophysis
 - in contact with colloid
 - *Follicles are delimited by **basement membrane** from external side*
 - **parafollicular cells (c cells)**
 - Produce :
 - **Calcitonin** :
 - decreases blood calcium levels (increase bone)
 - Regulated by blood calcium levels
 - not in contact with colloid
- **stroma**
 - Interstitial connective tissue
 - Many fenestrated blood capillaries

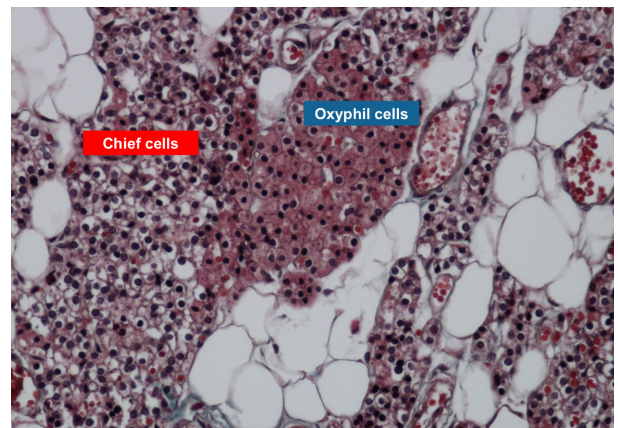
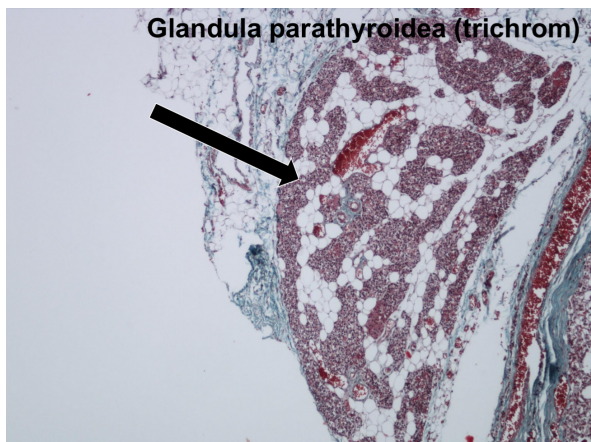


PARATHYROID GLAND

- 2 pairs of small endocrine bodies on the posterior surface of the thyroid gland
- Superior and inferior
- Trabecular type of gland

microscopic structure

- **Capsule**
 - Thin loose CT capsule
 - Divides glands into lobules and separates chords
 - Separates parathyroid from thyroid gland
- **stroma**
 - Interstitial connective tissue containing blood and lymph
 - Lipid droplets
- **parenchyma** :
 - 2 cell types arranged into chords
 - **Cells**
 - **Chief cells (principal / parathyroid)**
 - Polygonal shape
 - Basophilic
 - Produce :
 - **parathyroid hormone**
 - Increases blood calcium level (stimulates osteoclasts)
 - Absorption of Ca in GI tract
 - Increase phosphates in urine
 - dense (dark) chief cells : active form, more secretory granules and less glycogen
 - Pale chief cells : inactive form, less secretory granules and more glycogen
 - **oxyphil cell**
 - Larger polygonal
 - Small amount found in groups or singly
 - Acidophilic
 - Many mitochondria
 - Unknown function

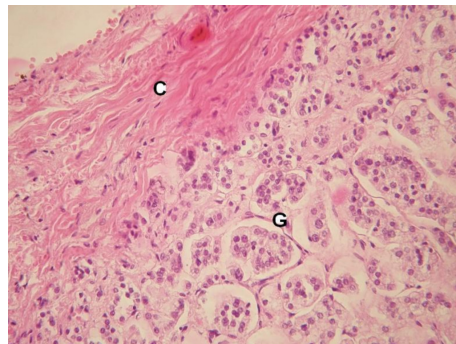


ADRENAL / SUPRARENAL GLAND

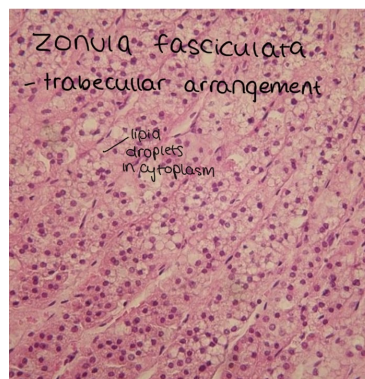
- Lie in superior pole of kidneys in renal adipose capsule
- Retroperitoneal
- Flattened structure
- Right : triangular shaped
- Left : semilunar shape
- Trabecular type of gland

Microscopic structure

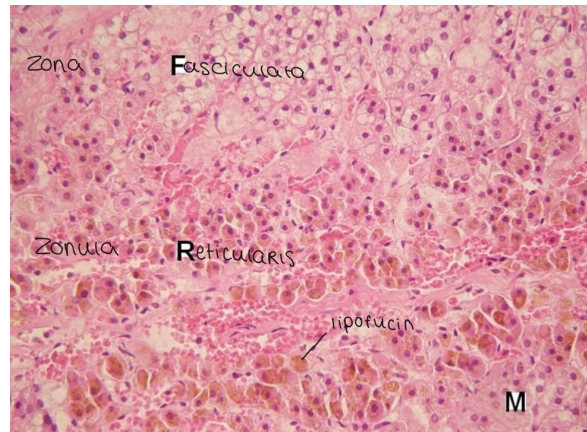
- capsule
 - Strong fibrous capsule with septa protruding from it
 - Septa : extend from capsule, carry blood vessels and nerves
- cortex
 - Most of the gland
 - Mesodermal mesenchyme
 - Cells of cortex produce **steroid hormones**
 - 3 layers :
 - Zonula glomerulosa
 - Grouped fascicles / arched clusters of cells
 - Cells :
 - Columnar - pyramidal
 - Small
 - Produce **mineralocorticoids - aldosterone**
 - **Regulates BP**



- Zonula fasciculata
 - Longitudinal trabeculae of cells separated by sinusoidal capillaries
 - Cells :
 - Large
 - Polyhedral
 - Lipid droplets in cytoplasm
 - Produce **glucocorticoids - cortisol + corticosterone**
 - Widest zone



- Zonula reticularis
 - **Anastomosing chords** separated by fenestrated capillaries
 - cells :
 - Smallest cells
 - Lipofucin pigment granules
 - Produce **androgens**



- medulla

- Cells produce **catecholamines**
- Neural crest cells migrate into developing gland
- Cells :
 - Chromafin cells
 - Arranged into **chords / fascicles**
 - “Altered neuron” : axons arrive to the cells
 - Produce **noradrenaline (norepinephrine)** which can be transformed to **adrenaline (epinephrine)**
 - Multipolar autonomic ganglion cells :
 - Neuron, axon runs to border of cortex
 - Helps regulate secretion of hormones

